

9 Infection with *Mycobacterium tuberculosis* results in tuberculosis (TB), which is one of the top ten causes of death in the world.

(a) In 2019, an estimated 1.2 million children, 3.2 million women and 5.6 million men developed TB.

Calculate the ratio of children to women to men who developed TB.

(1)

Answer

(b) Explain how infection with *Mycobacterium tuberculosis* may result in death.

(3)

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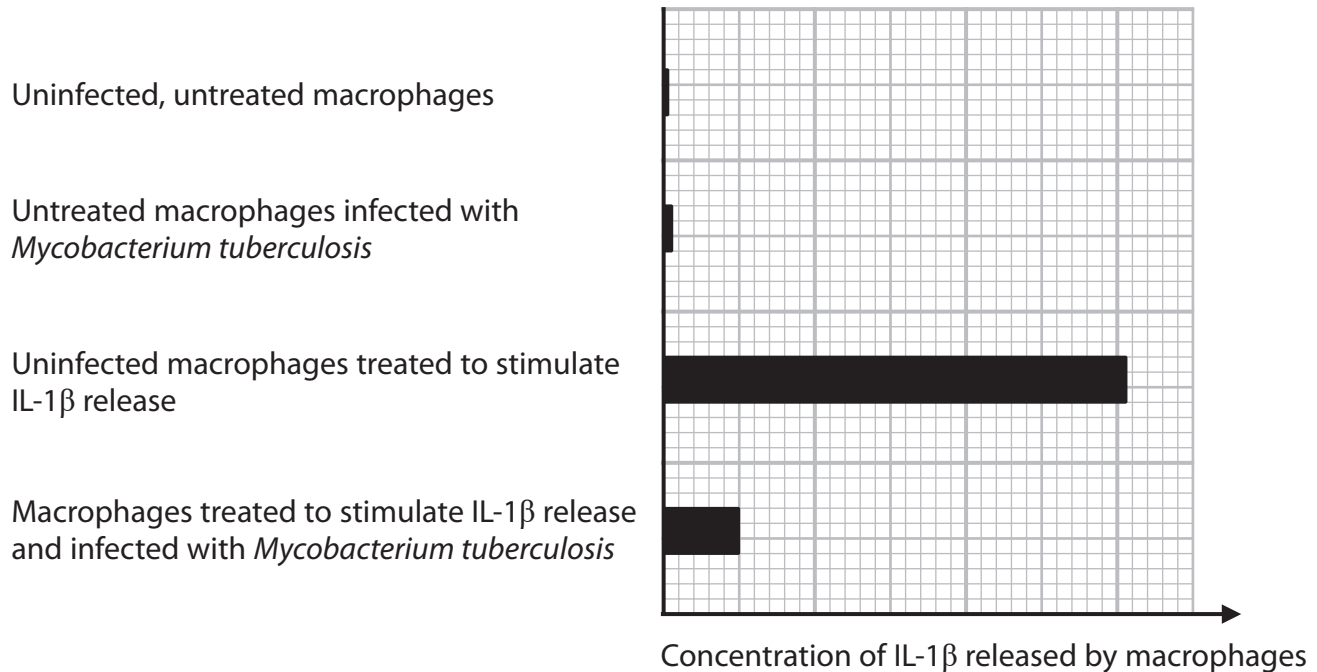
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- (c) *Mycobacterium tuberculosis* has evolved to evade host immunity by interfering with macrophage function.

In one investigation, scientists showed that *Mycobacterium tuberculosis* can inhibit the activation of a process in macrophages. This inhibition results in less of the chemical IL-1 β being released.

- (i) The graph shows some of the results of this investigation.



Explain why the macrophages had to be treated to stimulate IL-1 β release in this investigation.

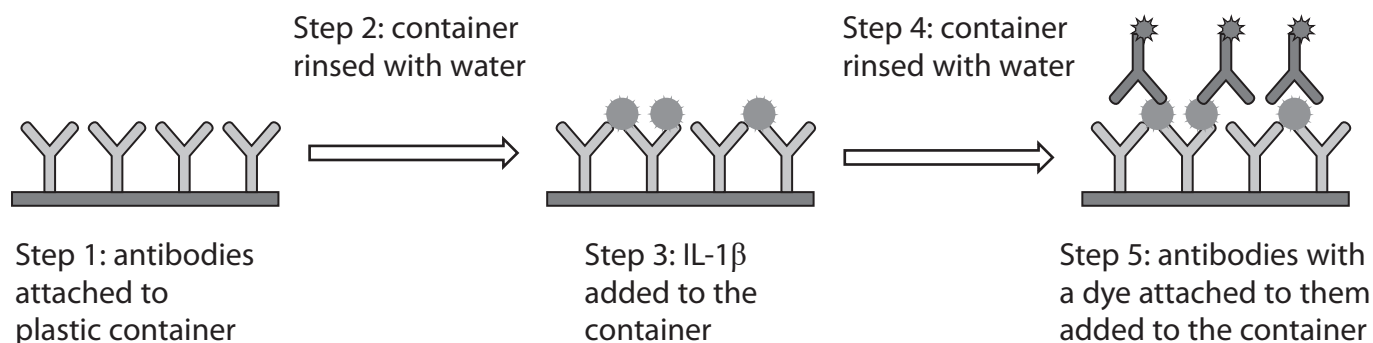
Use the information in the graph to support your answer.

(2)

(ii) An ELISA is used to detect a specific antigen.

The IL-1 β was detected using an ELISA.

The diagram shows an ELISA.



Key



first antibody used

IL-1 β



antibody with dye attached

Explain how the steps in the ELISA result in the identification of IL-1 β .

Use the information in the diagram to support your answer.

(4)

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(4)

(1)

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TOTAL FOR PAPER = 90 MARKS